Your grade: 93.33%

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Next item →

1. To help you practice strategies for machine learning, this week we'll present another scenario and ask how you would act. We think this "simulator" of working in a machine learning project will give you an idea of what leading a machine learning project could be like!

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You are employed by a startup building self-driving cars. You are in charge of detecting road signs (stop sign, pedestrian crossing sign, construction ahead sign) and traffic signals (red and green lights) in images. The goal is to recognize which of these objects appear in each image. As an example, this image contains a pedestrian crossing sign and red traffic lights.

$$y^{(i)} = \begin{array}{cl} \left[\begin{array}{c} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{array}\right] & \begin{array}{l} \text{“stop sign”} \\ \text{“pedestrian crossing sign”} \\ \text{“construction ahead sign”} \\ \text{“red traffic light”} \\ \text{“green traffic light”} \end{array} \end{array}$$

Your 100,000 labeled images are taken using the front-facing camera of your car. This is also the distribution of data you care most about doing well on. You think you might be able to get a much larger dataset off theinternet, which could be helpful for training even if the distribution of internet data is not the same.

Suppose that you came from working with a project for human detection in city parks, so you know that detecting humans in diverse environments can be a difficult problem.

What is the first thing you do?

Assume each of the steps below would take about an equal amount of time (a few days).

- ○ Spend a few days collecting more data to determine how hard it will be to include more pedestrians in your dataset.
- ◉ Train a basic model and proceed with error analysis.
- ○ Start by solving pedestrian detection, since you already have the experience to do this.
- ○ Leave aside the pedestrian detection, to move faster and then later solve the pedestrian problem alone.

✓ Nice work

It's most efficient to create a basic system and iterate based on what the errors reveal.

2. Your goal is to detect road signs (stop sign, pedestrian crossing sign, construction ahead sign) and traffic signals (red and green lights) in images. The goal is to recognize which of these objects appear in each image. You plan to use a deep neural network with ReLU units in the hidden layers.

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True or False: For the output layer, a softmax activation would be a good choice because this is a multi-task learning problem.

- True
- False

Nice work

Softmax would be a good choice if one and only one of the possibilities (stop sign, speed bump, pedestrian crossing, green light and red light) was present in each image. However, in this case, multiple objects can be present in a single image. Therefore, a sigmoid activation for each output node is more appropriate.

3. **True or False:** When trying to determine what strategy to implement to improve the performance of a model, you manually check all images of the training set where the algorithm was successful.

1 / 1 point

- False
- True

Nice work

The training set is typically very large, and manually checking all the successful images would be time-consuming and inefficient. It's more effective to focus on the errors in the dev set to understand where the model needs improvement.

4. After working on the data for several weeks, your team ends up with the following data:

1 / 1 point

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1 / 1 point

- 100,000 labeled images taken using the front-facing camera of your car.
- 900,000 labeled images of roads downloaded from the internet.
- Each image's labels precisely indicate the presence of any specific road signs and traffic signals or combinations of them. For example, $y^{(i)} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ means the image contains a stop sign and a red traffic light.

True or False: In multi-task learning, if some examples have missing labels (for example: $\begin{bmatrix} 0 \\ ? \\ 1 \\ 1 \\ ? \end{bmatrix}$), the learning algorithm **cannot** use those examples.

- True
- False

Nice work

As seen in the lecture on multi-task learning, you can compute the cost function in a way that it is not affected by missing labels. The algorithm can still learn from the available labels in the example.

5. **True or False:** The distribution of data you care about contains images from your car's front-facing camera, which comes from a different distribution than the images you were able to find and download off the internet. The best way to split the data is using the 900,000 internet images to train, and divide the 100,000 images from your car's front-facing camera between dev and test sets.

1 / 1 point

5. **True or False:** The distribution of data you care about contains images from your car's front-facing camera, which comes from a different distribution than the images you were able to find and download off the internet. The best way to split the data is using the 900,000 internet images to train, and divide the 100,000 images from your car's front-facing camera between dev and test sets.

1 / 1 point

- True
- False

Nice work

100,000 images are too many to use in dev and test. A better distribution would be to use 80,000 of those images to train, and split the rest between dev and test. This ensures that your dev and test sets reflect the "real-world" data you care about.

6. Assume you've finally chosen the following split between the data:

1 / 1 point

Dataset:	**Contains:**	**Error of the algorithm:**
Training	940,000 images randomly picked from (900,000 internet images + 60,000 car's front-facing camera images)	1%
Training-Dev	20,000 images randomly picked from (900,000 internet images + 60,000 car's front-facing camera images)	5.1%
Dev	20,000 images from your car's front-facing camera	5.6%
Test	20,000 images from the car's front-facing camera	6.8%

You also know that human-level error on the road sign and traffic signals classification task is around 0.5%. Which of the following is true?

- You have a high variance problem.
- You have a large data-mismatch problem.
- The size of the train-dev set is too large.
- You have a high bias.

Nice work

The large difference between the training-dev error and the training error indicates that the model is not generalizing well to unseen data from the same distribution, which is a sign of high variance.

7. Assume you've finally chosen the following split between the data:

1 / 1 point

Dataset:	Contains:	Error of the algorithm:
Training	940,000 images randomly picked from (900,000 internet images + 60,000 car's front-facing camera images)	8.8%
Training-Dev	20,000 images randomly picked from (900,000 internet images + 60,000 car's front-facing camera images)	9.1%
Dev	20,000 images from your car's front-facing camera	14.3%
Test	20,000 images from the car's front-facing camera	14.8%

Human-level error on this task is approximately 0.5%. (Bayes error is the lowest possible error rate for a task. Human-level error is a good estimation of Bayes error.)

A friend believes the mixed internet/car images (Training) have a lower Bayes error than the car camera images (Dev/Test). What do you think?

- Your friend is likely incorrect.
- Your friend is likely correct.
- There's insufficient information to determine if your friend is correct or not.

Nice work

To accurately determine and compare Bayes errors, we need to compare the human-level error for each data distribution. The overall human level error does not provide enough information to compare the two distributions.

the two distributions.

8. **You decide to focus on the dev set and check by hand what the errors are due to. Here is a table summarizing your discoveries:**

1 / 1 point

Overall dev set error	15.3%
Errors due to incorrectly labeled data	4.1%
Errors due to foggy pictures	3.0%
Errors due to partially occluded elements.	7.2%
Errors due to other causes	1.0%

In this table, 4.1%, 7.2%, etc. are a fraction of the total dev set (not just examples of your algorithm mislabeled). For example, about 7.2/15.3 = 47% of your errors are due to partially occluded elements.

True or False: You shouldn't invest all your efforts to get more images with partially occluded elements since 4.1 + 3.0 + 1.0 = 8.1 > 7.2.

- (selected) False
- True

Nice work

While the sum of other errors is greater, focusing solely on the numbers ignores crucial factors like the cost and accessibility of acquiring more images with partially occluded.

9. **You can buy a specially designed windshield wiper that helps wipe off some of the raindrops on the front-facing camera.**

1 / 1 point

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1 / 1 point

Overall dev set error	**15.3%**
Errors due to incorrectly labeled data	**4.1%**
Errors due to foggy pictures	**8.0%**
Errors due to rain drops stuck on your car's front-facing camera	**2.2%**
Errors due to other causes	**1.0%**

Which one of the following statements do you agree with?

- (selected) 2.2% would be a reasonable estimate of the maximum amount this windshield wiper could improve performance.
- 2.2% would be a reasonable estimate of the minimum amount this windshield wiper could improve performance.
- 2.2% would be a reasonable estimate of how much this windshield wiper will improve performance.
- 2.2% would be a reasonable estimate of how much this windshield wiper could worsen performance in the worst case.

Nice work

You will probably not improve performance by more than 2.2% by solving the raindrops problem.

10. **You decide to use data augmentation to address foggy images. You find 1,000 pictures of fog off the internet and "add" them to clean images to synthesize foggy days, like this:**

1 / 1 point

Nice work

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10. You decide to use data augmentation to address foggy images. You find 1,000 pictures of fog off the internet and "add" them to clean images to synthesize foggy days, like this:

1 / 1 point

Which one of the following do you agree with?

- ○ If used, the synthetic data should be added to the training/dev/test sets in equal proportions.
- ◉ If used, the synthetic data should be added to the training set.
- ○ With this technique, we duplicate the size of the training set by synthesizing a new foggy image for each image in the training set.
- ○ It is irrelevant how the resulting foggy images are perceived by the human eye; the most important thing is that they are correctly synthesized.

Nice work

The synthetic data can help train the model to get better performance on the dev set, but it shouldn't be added to the dev or test sets because they don't represent our target in a completely accurate way.

11. **After working further on the problem, you've decided to correct the incorrectly labeled data on the dev set. Which of these statements do you agree with?**

1 / 1 point

- [x] You do not necessarily need to fix the incorrectly labeled data in the training set because it's acceptable for the training set distribution to differ from the dev and test sets. Note that it is important that the dev set and test set have the same distribution.

Nice work

True, deep learning algorithms are quite robust to having slightly different train and dev distributions.

- [x] You should also correct the incorrectly labeled data in the test set, so that the dev and test sets continue to come from the same distribution.

Nice work

Yes, because you want to ensure that your dev and test data come from the same distribution for your algorithm to make your team's iterative development process efficient.

- [ ] You should not correct the incorrectly labeled data in the test set, so that the dev and test sets continue to come from the same distribution.
- [ ] You should correct incorrectly labeled data in the training set as well to avoid your training set now being even more different from your dev set.

12. **So far, your algorithm only recognizes red and green traffic lights. One of your colleagues in the startup is starting to work on recognizing a yellow traffic light. Some countries refer to it as an orange light; however, we will use the US convention of calling it yellow. Images containing yellow lights are quite rare, and she doesn't have enough data to build a good model. She hopes you can help her out using transfer learning.**

0 / 1 point

What do you tell your colleague?

- ( ) She should try using weights pre-trained on your dataset and fine-tuning further with the yellow-light

12. **So far, your algorithm only recognizes red and green traffic lights. One of your colleagues in the startup is starting to work on recognizing a yellow traffic light. Some countries refer to it as an orange light; however, we will use the US convention of calling it yellow. Images containing yellow lights are quite rare, and she doesn't have enough data to build a good model. She hopes you can help her out using transfer learning.**

0 / 1 point

What do you tell your colleague?

- She should try using weights pre-trained on your dataset and fine-tuning further with the yellow-light dataset.
- If she has (say) 10,000 images of yellow lights, randomly sample 10,000 images from your dataset and put your and her data together. This prevents your dataset from "swamping" the yellow lights dataset.
- You cannot help her because the distribution of data you have is different from her's and is also lacking the yellow label.
- Recommend that she try multi-task learning instead of transfer learning using all the data.

Try again

Multi-task learning is not the most suitable approach here, as transfer learning is more effective when dealing with a small dataset and leveraging a pre-trained model.

13. **One of your colleagues at the startup is starting a project to classify road signs as stop, dangerous curve, construction ahead, dead-end, and speed limit signs. He has approximately 30,000 examples of each image and 30,000 images without a sign.**

1 / 1 point

True or False: This case could benefit from using multi-task learning.

- False
- True

Nice work

Multi-task learning is suitable here due to the shared high-level features among the required road signs.

14. You want to recognize red and green lights in images. You have two approaches:

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- **Approach 1**: Input an image (x) into a neural network that directly predicts whether a red or green light is present (y).
- **Approach 2**: First, detect the traffic light in the image (if any). Then, determine the color of the illuminated lamp.

Which approach is a better example of an end-to-end approach?

- (selected) Approach 1
- Approach 2

Nice work

Approach 1 directly maps the input (x) to the output (y) in a single step, which is the definition of an end-to-end approach.

15. To recognize a stop sign, you use the following approach:

1 / 1 point

First, localize any traffic sign in an image. **After that**, determine if the sign is a stop sign or not.

This is a better approach than an end-to-end model for which of the following cases? **Choose the best answer.**

- There are available models which we can use to transfer knowledge.
- There is a large amount of data.
- (selected) There is not enough data to train a big neural network.
- The problem has a high Bayes error.

Nice work

When data is limited, a two-step approach can be more effective than training a large end-to-end model.